

Name:

Roll No:

INSTITUTE OF BUSINESS ADMINISTRATION
Department of Computer Science, Karachi
Course: Introduction to Artificial Intelligence (CSE-307)
Instructor: Dr. Syed Ali Raza, Sumaira Saeed, and Dr. Tahir Syed
Mid-Term Exam – Spring 2025

Duration: 2 hours

Total marks: 54

Instructions: Provide answers in the space provided after each question. You may use an extra sheet for rough work, but not for providing answers. Please answer using a pen only (pencils are not permitted).

UNINFORMED SEARCH

- 1) [6 marks] Consider the grid shown in Figure 1, where each cell represents a node and is named. Apply Depth-First Search (DFS) to find a path from the starting node A to the goal node B. Movement Rules:

- From each node, four moves are possible: Up, Right, Left, and Down.
- Moves are explored in the order: Up, Right, Left, Down.
- Moving outside the grid boundaries (the "walls") results in staying in the same node.
- To avoid loops, ignore nodes that are already in the current path.

B	1	2	3
4	5	6	7
8	9	10	11
12	13	A	14

Figure 1

Write the sequence of nodes explored by DFS.

Answer:

Write the total number of nodes expanded in the search tree (including A and B).

Answer:

Write the sequence of nodes in the path found by DFS.

Answer:

- 2) [2 marks] Consider Uniform Cost Search (UCS) and Breadth-First Search (BFS). Assume all edge costs are non-negative.

Define the term 'effective depth' in the context of a search problem with varying edge costs.

Answer:

Under what specific condition, related to the edge costs, does the time complexity of UCS (expressed in Big Oh notation) become equivalent to the time complexity of BFS, when considering the 'effective depth'?

Answer:

- 3) [4 marks] Consider the graph shown in Figure 2. Apply UCS to find the shortest path from the start state A to the goal state E. Assume all edge costs are non-negative. To avoid loops, ignore nodes that are already in the current path.

Write the sequence of nodes explored by UCS.

Answer:

Write the sequence of nodes in the shortest path found by UCS.

Answer:

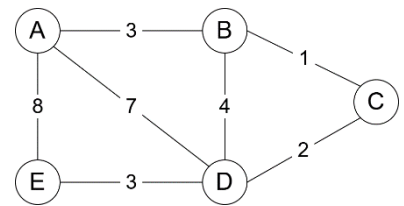


Figure 2

INFORMED SEARCH

- 4) [3 marks] Consider a 3-puzzle problem (a modified version of 8-puzzle). The A* algorithm has found the shortest path passing through three states: the start state, an intermediate state, and the goal state. The heuristic used by A* is the sum of Manhattan distances for each numbered tile (i.e., the number of horizontal and vertical moves needed to place each piece in its goal position). The algorithm ignores the already visited states. Note, the blank space can be moved up, down, left, or right.

In the empty space below, write the intermediate state found by the A* and its heuristic value.

Start state:

	2
1	3

$h = 2$

Intermediate state:

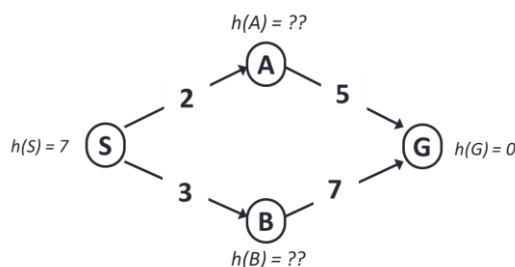
$h = \underline{\hspace{2cm}}$

Goal state:

1	2
3	

$h = 0$

- 1) [5 marks] In A* search, the algorithm terminates when the goal state is dequeued. Consider the following graph, where S is the start state and G is the goal state. The edge weights represent the cost of traversing between states. Determine five possible combinations of admissible heuristic values for states A ($h(A)$) and B ($h(B)$) that would cause A* to return a suboptimal path (a path with a higher cost than the shortest path) if the algorithm terminated when the goal state G was enqueued. Provide your answer in the table below.



No.	$h(A)$	$h(B)$
1		
2		
3		
4		
5		

- 2) [3 marks] In A* search, suppose we can modify the evaluation function to assign different weights to the past cost ($g(n)$) and the heuristic cost ($h(n)$) using the formula: $f(n) = w_1 * g(n) + w_2 * h(n)$. Assume $h(n)$ is originally an admissible heuristic.

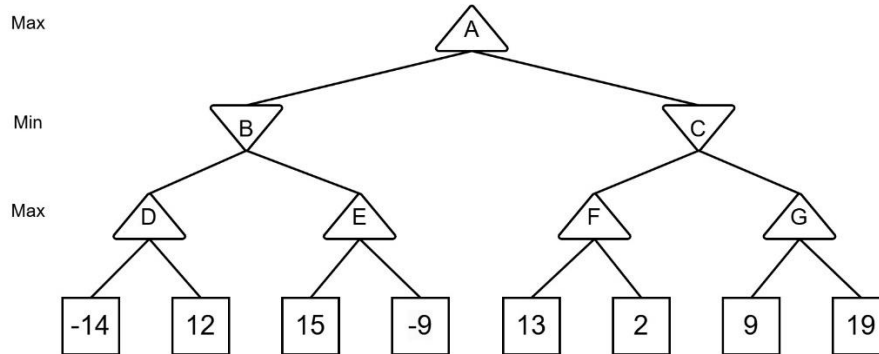
Consider the case where $w_1 = 1$. For each of the following scenarios for w_2 , determine whether the resulting heuristic in the modified $f(n)$ function will remain admissible or become inadmissible.

Scenarios	Answers
$w_2 > 1$	

$w_2 = 1$	
$0 < w_2 < 1$	

ADVERSARIAL SEARCH

- 3) [7 marks] Consider the following game tree. Apply the Minimax algorithm with alpha-beta pruning and answer the following questions:



Questions	Answers
What is the final value of node A?	
What is the final Alpha at node A?	
What is the final Beta at node A?	
What is the final value of node B?	
What is the final Alpha at node C?	
Write the names of all the nodes and their left or right branch that are pruned. E.g., if the right branch of node D is pruned, then write "D, right".	

- 4) [6 marks] Answer the following questions about the Minimax algorithm. Suppose 'm' represents the maximum depth, and 'b' represents the possible actions at each state.

Questions	Answers
What is the time complexity of Minimax?	
What is the space complexity of Minimax?	
What is the time complexity of Minimax with alpha-beta pruning with ideal ordering moves?	
What is the time complexity of Minimax with alpha-beta pruning with worst-case ordering moves?	
In some cases, the alpha-beta pruning can provide results different from the full Minimax results (True or False)?	
What is a single level of choice in the game tree called?	

LOCAL SEARCH

- 5) [1 mark] Name the local search algorithms which operate by maintaining only a single current state in memory?

Answer: _____

- 6) [1 mark] In Simulated Annealing, if the temperature is held sufficiently high (to allow almost all moves to be accepted) for the first 99 iterations and then dropped to zero on the 100th iteration, will the algorithm behave as a global search or a local search algorithm?

Answer: _____

- 7) [4 marks] Apply the Simulated Annealing (SA) algorithm to a maximization problem with two variables x_1 and x_2 with values in the range -10 to +10. The evaluation function is $f(x_1, x_2) = x_2^3 - x_1^2 + 12$. The following table shows the first two steps of the algorithm with some missing values. The table provides the initial values of the two variables, the actions applied to each variable in each step, and the random numbers for each step (if needed). Note: An action is applied by adding or subtracting 1 from each variable's value. Your task is to complete the first two steps of SA and answer the following questions (no need to fill in the missing values in the table). This formula may help you, $e^{\Delta E/T}$.

Step	Temperature	x_1	x_2	Action x_1	Action x_2	Next x_1	Next x_2	Evaluation	Next Evaluation	Acceptance Prob	Random Number	Decision
1	50	5	5	1	-1						0.2939	
2	40			-1	-1						0.2858	

Questions	Answers
What is the acceptance probability in step 1 (if applied)?	
What is the decision for step 1 (accept or reject)?	
What is the acceptance probability in step 2 (if applied)?	
What is the decision for step 2 (accept or reject)?	

EVOLUTIONARY COMPUTATION

- 8) [1 mark] Which operation in an Evolutionary Algorithm is primarily responsible for introducing randomness and disrupting the focused search for neighboring solutions?

Answer: _____

- 9) [6 marks] A logistics company is optimizing delivery routes for five drivers. Each driver follows a route covering four locations (X, Y, Z, and W) in a specific order. The total delivery time (in minutes) for each route is used as the fitness value (lower time is better). The goal is to evolve better routes with less total delivery time using an Evolutionary Algorithm (EA). The distance (time) between cities is given in the table below:

X-Y	X-Z	X-W	Y-Z	Y-W	Z-W
10	15	20	25	30	35

Complete the table below that shows the initial population (routes and their total delivery times):

Individual (Driver)	Route (Order of Locations)	Total Delivery Time (minutes)
A	X → Y → Z → W	
B	Y → W → X → Z	
C	Z → X → W → Y	
D	W → Y → X → Z	
E	X → Z → W → Y	

Perform one complete iteration of an Evolutionary Algorithm using the following steps: use binary tournament selection, but follow this predefined pairing and fill the table to specify selected parents:

Tournament Round	Competing Routes	Selected Winner (Better Route)
1	A , B	Parent 1 =
2	C , D	Parent 2 =

Perform one-point crossover at a fixed position after the second location (crossover point = after the 2nd location). One-point crossover selects a single crossover point on the parent chromosomes and swaps all genes after that point between the two parents to create offspring.

Offspring 1 = ____ → ____ → ____ → ____
Offspring 2 = ____ → ____ → ____ → ____

Apply swap mutation, where the 2nd and 4th locations swap places.

Offspring 1 (after mutation) = ____ → ____ → ____ → ____
Offspring 2 (after mutation) = ____ → ____ → ____ → ____

Keep the five best performing individuals (with the lowest delivery times). Show new population below:

Individual (Driver)	Route (Order of Locations)	Total Delivery Time (minutes)

Analyze whether the population is moving toward a better solution (lower delivery time). Suggest additional steps to further optimize the routes.

Answer: _____

PARTICLE SWARM OPTIMIZATION

10) [5 marks] Apply Particle Swarm Optimization (PSO) to maximize the function,

$$f(x) = -x^2 - 3x - 20$$

Use 4 particles with the initial positions and velocities provided in the table below.

Assume, the inertial weight (w) is 1, C1 = 0.5 and C2=0.5. The random numbers are R1 = 0.213 and R2= 0.876.

For all the calculations, use numbers to three decimal places.

The table has other columns with missing values. Your task is to complete one iteration of PSO and find the missing values to answer the following questions. No need to fill in the missing values in the table, just answer the questions.

Particle No.	Initial Positions	Initial Velocity	Initial Fitness	Initial pBest	Initial gBest	New Velocity	New position	New fitness	updated pBest	updated gBest
P1	-0.8	0.5								
P2	1	-0.5								
P3	-2	0.5								
P4	1.5	0								

Questions	Answers
What are the updated values of pBest for each particle?	Updated pBest for P1:
	Updated pBest for P2:
	Updated pBest for P3:
	Updated pBest for P4:
What is the updated value of gBest?	Updated gBest: